COMPARATIVE GENE VIEWER

OFFICIAL MANUAL

CGV User Manual

Web and Desktop Edition

Search | Compare | Interpret | Export

A complete operational and interpretive reference for gene-centered visualization, comparative analysis, functional context, and publication output.

VERSION	REVISION	APPLICATION
1.1	30 July 2026	cgv.mobilomics.org

Contents

About this manual

Purpose

This manual is written for researchers, students, instructors, and technical users who want to use CGV without first learning its implementation details. It explains what each major control does, when to use it, how to interpret the result, and how to preserve or export the analysis.

The procedures apply to both CGV Web and CGV Desktop unless a section is marked **Desktop only**. The scientific interface is shared by both editions. Desktop adds a private local runtime, persistent local data storage, an organism catalog, and operating-system integration.

Product scope

CGV is designed for gene-centered structural and comparative analysis. It is especially useful for:

- comparing several annotated genes within one organism;
- comparing one gene across several organisms;
- inspecting gene, transcript, exon, CDS, UTR, intron, and neighboring-gene structure;
- comparing transcript isoforms and cross-species structural relationships;
- examining local sequence conservation with aligned synteny, LASTZ blocks, and MultiPIP;
- reviewing gene metrics, sequence composition, Gene Ontology annotations, functional summaries, literature, and protein interaction context;
- exporting figures, tables, sequences, and reproducible work sessions;
- sharing immutable interactive read-only reports by secret URL in CGV Web or self-contained HTML in CGV Desktop.

CGV is not a replacement for differential-expression analysis, variant calling, genome assembly, whole-genome browsing, phylogenetic inference, population-genetic analysis, protein 3D prediction, or clinical interpretation.

> **IMPORTANT** CGV displays and analyzes the annotations and assemblies supplied by the selected data source. A missing feature can reflect the source annotation rather than the biological absence of that feature.

Conventions

- **Bold text** identifies labels that appear in the CGV interface.
- `Monospace text` identifies file types, file names, identifiers, or literal values.
- Numbered steps are actions to perform in order.
- A note explains behavior that affects interpretation or reproducibility.
- A tip recommends an efficient working practice.
- Organism names are written using their scientific names where possible.

Terms used throughout the manual

Multi-Gene Search means several gene queries within one selected organism. **Cross-Species Gene Search** means one target gene resolved across several selected organisms. A **canonical card** is the primary displayed transcript record for a gene. An **isoform card** is an additional transcript record revealed from the canonical card. A **result card** contains a header, an interactive gene model, and a footer with statistics and analysis actions.

1. CGV at a glance

CGV follows a consistent analysis path regardless of the data source or search mode.

THE CGV ANALYSIS PATH

1	2	3	4	5
SELECT WORKFLOW	PREPARE DATA	GENERATE	ANALYZE	PRESERVE
One organism with many genes, or many organisms with one gene.	Use preloaded data, NCBI Search, uploaded files, or mixed sources.	Resolve identifiers and build the interactive result cards.	Inspect structure, alignments, metrics, function, context, and literature.	Export SVG, PNG, CSV, FASTA, and the complete RDS session.

Verify source provenance and the resolved gene or transcript before interpreting differences.

Choose the correct search mode

Research question	Use	Query pattern	Typical result
How do several genes differ in one organism?	Multi-Gene Search	Multiple genes, one organism	One or more gene and transcript cards per query
How do transcript isoforms of one gene differ?	Multi-Gene Search	One gene with multiple transcripts	Canonical card, expandable isoform cards, transcript alignment
How does one gene compare across species?	Cross-Species Gene Search	One gene, multiple organisms	Representative cards by organism and comparative views
Where are conserved local sequence blocks?	Cross-Species Gene Search	One gene, multiple genomes	LASTZ Blocks or MultiPIP
Do I need every family member in one organism?	Multi-Gene Search	Add the family members as a batch	Within-organism family comparison

NOTE Cross-Species Gene Search resolves the selected target across organisms. It is not a complete inventory of every related family member present in each organism.

The first ten minutes

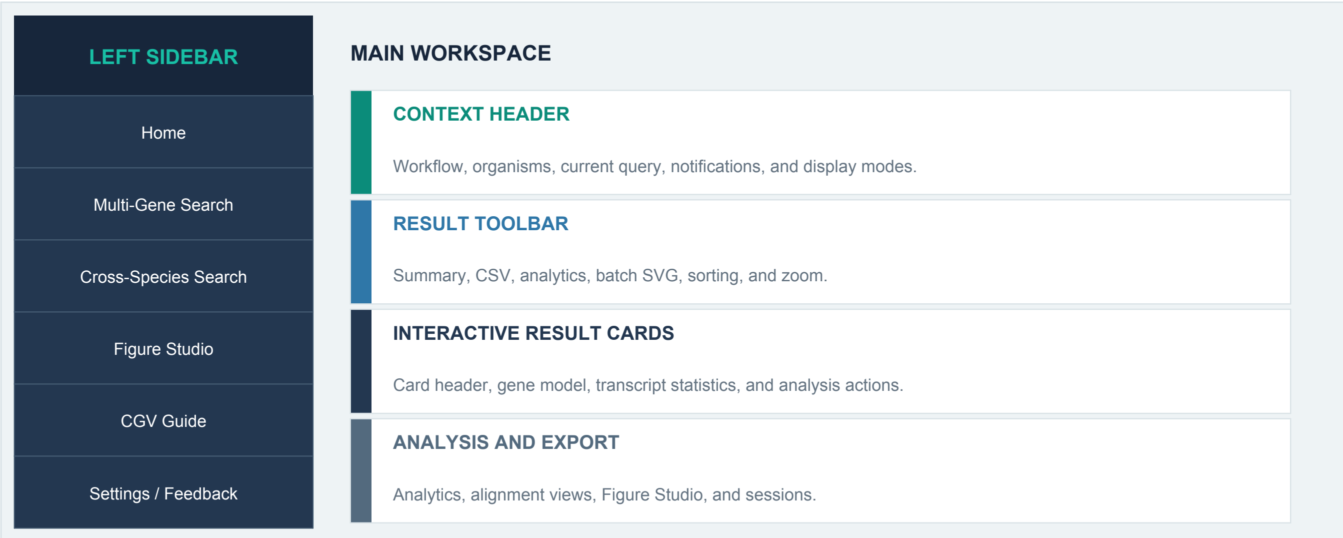
1. Open **Multi-Gene Search** or **Cross-Species Gene Search** from the left sidebar.
2. Select a data source: **Preloaded organism**, **NCBI Search**, **Upload files**, or, in Cross-Species Search, **Mixed sources**.
3. Select the organism or organisms.
4. Enter a gene symbol, stable identifier, or recognized alias in the sidebar search.
5. In Multi-Gene Search, add more genes to the batch if required.
6. Select **Generate visualization**.
7. Inspect the result cards in **Compact** or **Detailed** view.
8. Open **Show Summary Table** and **Show Analytics** for structured comparisons.
9. Use card actions for function, network, GO, external records, charts, literature, or sequence download.
10. Export the required figures and tables, save the work session from **Settings**, or use **Share** to create an interactive report for other readers.

A recommended working pattern

- Begin with **Compact** view to verify that the intended genes and organisms were resolved.

- Expand transcript isoforms only for genes that require transcript-level interpretation.
- Switch to **Detailed** view before documenting feature-level conclusions.
- Use alignment views after verifying the underlying transcript structures.
- Save a session before making a major source, organism, or layout change.
- Export SVG for editable publication graphics and CSV for auditable supporting data.

2. Interface tour



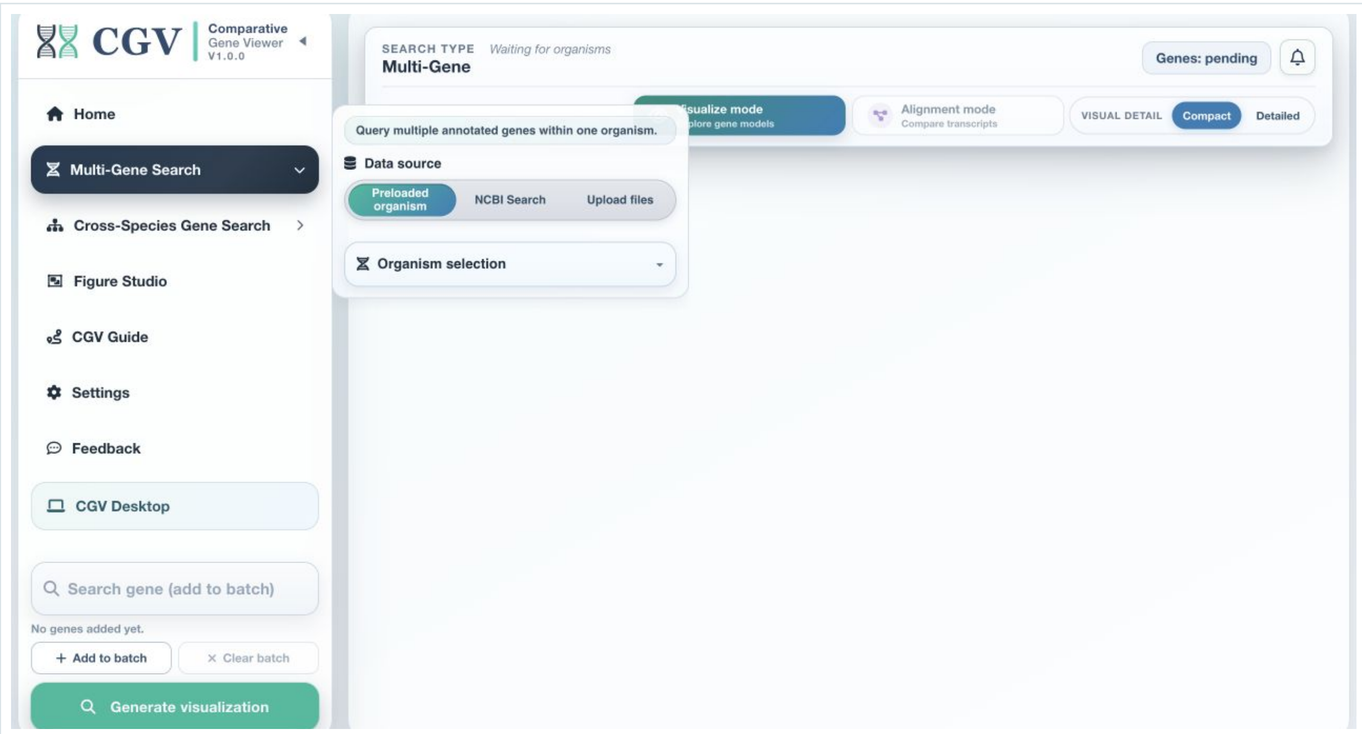


Figure 2.1. The real CGV workspace with the Multi-Gene configuration panel open. The sidebar contains workflow navigation and gene entry; the main header controls visualization mode and detail.

2.1 Left sidebar

The sidebar is the primary navigation and search surface.

Item	Purpose
Home	Product overview, workflow orientation, scope, troubleshooting, and FAQ
Multi-Gene Search	Configure one organism and one or more gene queries
Cross-Species Gene Search	Configure several organisms and one shared gene query
Figure Studio	Assemble available CGV charts into a multi-panel publication figure
CGV Guide	Follow the video-based step-by-step learning routes
Settings	Configure appearance, deletion confirmation, quick actions, alias sources, organisms, and sessions
Feedback	Submit a bug report or feature suggestion
CGV Desktop	Review desktop capabilities and obtain the appropriate installer

Select the triangle control in the CGV brand area to collapse or expand the sidebar. The collapsed sidebar keeps the main navigation accessible and provides a compact gene-search panel.

2.2 Workflow configuration panels

Selecting a search workflow expands its configuration panel.

Multi-Gene Search contains:

- a data-source selector;
- a single-organism selector or source-specific controls;
- annotation and genome uploads;
- optional assembly-report, assembly-statistics, and GO-annotation uploads;
- NCBI organism search when that source is selected.

Cross-Species Gene Search contains:

- a data-source selector with **Mixed sources**;
- a multiple-organism selector;
- multi-file annotation and genome uploads;
- optional assembly and GO files for multiple organisms;
- an NCBI search queue for adding several assemblies.

2.3 Global search and gene batches

The search box changes its meaning with the active workflow:

- in Multi-Gene Search it accepts one gene or a batch;
- in Cross-Species Gene Search it focuses on one target gene.

Autocomplete suggestions are drawn from the selected local annotations. Choose a suggestion when possible to preserve the exact identifier spelling used by the source.

To build a Multi-Gene batch:

1. Type a gene.
2. Press **Space** or **Ctrl+Enter**, or select the add control.
3. Repeat for each gene.
4. Review the chips in the batch list.
5. Remove an individual chip if necessary, or use **Clear** to reset the batch.
6. Select **Generate visualization**.

CGV keeps unique entries and limits a single submitted batch to 24 genes for a stable interactive session. If a larger list is supplied, the first 24 unique genes are used.

2.4 Notification history and progress

The bell in the context header opens notification history. It collects search, download, alignment, warning, and completion messages that may otherwise disappear after a short notification. Use **Clear** in the status popup to empty the history.

Long operations display a working state. The interface remains usable while background alias resolution, cache preparation, deferred plot work, or supported external lookups complete.

LASTZ and MultiPIP runs, including automatic reference scoring, open the status popup automatically with a patience notice, current progress, and elapsed time. Keep CGV open until the run finishes. The transient working popup closes when the active run ends, and the completion or failure notice remains available in notification history.

2.5 Result context header

After data are loaded, the context header identifies:

- the active search type;
- selected organisms;
- the current gene or gene batch;
- whether CGV is in visualization or alignment mode;
- the active visual detail level.

The header provides **Visualize mode** and **Alignment mode**. Under Visualize mode, choose **Compact** or **Detailed**. Alignment choices depend on the workflow and loaded data.

Visualize mode also provides three independent context controls:

- **Scale** shows or hides genomic coordinates and the compressed

neighbor-distance scale;

- **Neighbors** shows or hides the nearest flanking genes;
- **Overlaps** shows or hides genes that overlap the displayed locus.

These controls change presentation only. Their state is retained for the current browser or desktop session and is applied to newly rendered cards.

2.6 Summary, analytics, sorting, and zoom

The toolbar above the result cards contains:

- **Show Summary Table** or **Hide Summary Table**;
- **Download CSV**;
- **Show Analytics** or **Hide Analytics**;
- batch SVG export controls;
- **Sort plots**;
- plot zoom controls.

Zoom changes the horizontal display scale of the interactive plot cards without changing the biological coordinates. Sorting changes presentation order, not the data or alignment calculation.

2.7 Share analysis

The unobtrusive **Share** action in the active result header appears after CGV has at least one result. In CGV Web it creates an expiring secret read-only URL. In CGV Desktop the same action prepares a self-contained interactive HTML file and reproducibility ZIP without uploading the analysis. See Sections 11.10 and 11.11 for privacy, contents, and reproducibility details.

2.8 Quick navigation button

When enabled in **Settings**, the floating quick-actions button provides:

- gene search and batch preview;
- Compact, Detailed, and alignment-mode switching;

- plot zoom;
- a scroll-to-top control when the page is away from the top.

This control is useful on long result pages. It does not replace the full workflow configuration panel.

3. Data sources and input requirements

3.1 Preloaded references

Preloaded references combine an indexed annotation with a matching genome assembly. They provide the fastest route to all CGV capabilities, including feature tooltips, sequence composition, FASTA export, promoter extraction, and local alignment.

In CGV Web, the server determines which references are available. In CGV Desktop, references installed from the organism catalog appear in the same **Preloaded organism** selectors.

Select the information icon beside an organism selection to review assembly metadata and statistics. Where organism imagery is available, CGV can also display licensed organism photographs with their source attribution.

3.2 NCBI Search

Use NCBI Search when the required assembly is not in the preloaded set.

1. Select **NCBI Search** as the data source.
2. Enter a scientific name, common name, or assembly-oriented search phrase.
3. Select the search icon.
4. Review matching assemblies and inspect the available accession, assembly name, level, and statistics.
5. Select the intended assembly.
6. Allow CGV to download and prepare the annotation, genome, assembly report, and assembly statistics.
7. Continue with the gene search after the source status reports readiness.

For Cross-Species Search, repeat the NCBI search to build the organism queue.

IMPORTANT	Different assemblies of the same organism may contain different coordinates, contig names, transcript versions, and annotation content. Record the assembly accession used in every reported result.

3.3 Upload files

Upload mode accepts user-supplied annotation and genome files.

Required files

Data	Accepted formats	Purpose
Annotation	`.gff`, `.gff3`, `.gtf`, `.txt`	Defines genes, transcripts, exons, CDS, UTR, and related features
Genome	`.fa`, `.fasta`, `.fna`, gzip-compressed FASTA, `.2bit`	Supports sequence composition, FASTA export, promoter extraction, and local alignment

Optional files

File	Accepted formats	Added capability
Assembly report	`.txt`	Assembly accessions, names, aliases, chromosome mapping, and metadata
Assembly statistics	`.txt`	Assembly-level size and contiguity statistics
GO annotations	`.gaf`, `.gaf.gz`	Local Gene Ontology lookup

Single-organism upload

For Multi-Gene Search, upload one annotation and one matching genome. Optional files must describe the same assembly.

Multi-organism upload

For Cross-Species Search, upload one annotation and one genome per organism. CGV pairs them by shared sequence identifiers and file-name evidence. Use clear, matching names and ensure chromosome or contig identifiers agree between each annotation and genome.

If CGV cannot establish a confident pairing, rename the files more clearly or verify that the genome build and sequence identifiers match.

3.4 Mixed sources

Mixed sources is available in Cross-Species Search. It combines:

- preloaded references;
- NCBI-prepared assemblies;
- uploaded annotation and genome pairs.

Use it when no single source contains the full comparison set. Every selected organism is treated as an independent track, while the result header records its source status.

3.5 What works when a file is missing

Available input	Expected behavior
Annotation and genome	Complete structural, sequence, promoter, alignment, and export workflow
Annotation only	Structural plots and annotation-derived metrics work; sequence-dependent values report N/A
Genome without annotation	Gene-centered plotting cannot begin because feature locations are not defined
Annotation and GO file	Local GO lookup is available
Annotation without GO file	GO can use the supported online fallback when a local record is unavailable
Assembly report and statistics	Assembly details and stronger chromosome naming are available

3.6 Data provenance checklist

Before reporting a result, record:

- organism and taxonomic identifier;
- assembly accession and version;
- annotation source and version;
- gene query and resolved local identifier;
- transcript identifier;
- whether an external alias was used;
- CGV search mode and visualization mode;
- alignment settings, when applicable;
- export date and CGV version.

4. Multi-Gene Search

4.1 When to use it

Use Multi-Gene Search when the comparison is centered on one organism. Suitable tasks include:

- comparing paralogs or members of a known family;
- comparing structural differences among selected genes;
- reviewing several loci from one experiment;
- examining alternative transcripts of one gene;

- aligning isoforms within one gene;
- comparing sequence and architecture metrics across a gene set.

4.2 Configure the source and organism

1. Select **Multi-Gene Search**.
2. Choose **Preloaded organism**, **NCBI Search**, or **Upload files**.
3. Select or prepare one organism.
4. Use the assembly information icon to confirm the expected build when precision matters.
5. Wait until the source panel reports readiness.

Changing organism after results exist prompts for confirmation because the current cards belong to the previous source. Save the session or export required results before replacing the source.

4.3 Enter one gene

1. Type a symbol, stable gene ID, transcript-oriented identifier, or alias.
2. Review autocomplete matches.
3. Select the exact record when it appears.
4. Select **Generate visualization** or press Enter.

CGV first searches the local annotation and local alias index. If the query is not an exact local match, it can offer partial-name suggestions or consult the external alias sources enabled in **Settings**.

4.4 Enter a gene batch

1. Type the first query.
2. Press Space or Ctrl+Enter to turn it into a chip.
3. Continue until the batch list contains the intended genes.
4. Review the batch count.
5. Select **Generate visualization**.

The search processes the batch in sequence and preserves successful plots. On completion, the notification history reports:

- genes plotted;
- genes not found;
- records without usable transcript structure;
- genes already plotted;
- search errors.

4.5 Resolve suggestions and ambiguous aliases

When no exact match exists, CGV can show similar local gene names. Multi-Gene Search allows one or more suggestions to be selected and plotted.

When an alias maps to several records, the **Several Alias Matches** dialog displays:

- the official symbol or local identifier;
- chromosome, coordinates, and strand;
- the alias or match role;
- the evidence source and confidence;
- a description when available;
- a recommended match when the evidence supports one.

Select only records that match the biological question. The **Alias** badge in a result card reopens the evidence used for the resolved record.

4.6 Review the first result

After the cards appear:

1. Confirm the organism, gene, transcript, and chromosome in the card header.
2. Hover the central gene model and its features.
3. Check strand direction and coordinates.
4. Review neighboring genes and the promoter-side indicator.
5. Check the footer for gene length, transcript length, sequence composition, and coordinates.
6. Expand the transcript control if multiple transcripts are available.

4.7 Choose a visual mode

Compact emphasizes the principal transcript structure and is best for dense comparisons.

Detailed exposes feature-level layers and richer tooltips. Use it to distinguish exon, CDS, UTR, intron, and other annotated elements.

Aligned compares transcripts belonging to one loaded gene. It becomes available when at least one selected gene has multiple transcript tracks.

LASTZ Blocks projects local sequence-alignment blocks over the canonical transcript locus.

MultiPIP stacks gap-free local segments by identity to show where conservation concentrates across the selected transcripts.

4.8 Sort and compare

Multi-Gene result cards can be sorted by:

- load order;
- total exon difference, high to low or low to high;
- transcript length;
- exon count;
- transcript name;
- chromosome.

Use the summary table for exact values and the cards for structural inspection. Sorting does not redefine the reference transcript or recalculate the underlying gene data.

4.9 Expand transcript isoforms

Canonical cards show a transcript-count control when additional isoforms are available.

1. Select the transcript-count control in the card footer.
2. Review the canonical copy and additional transcript cards shown below the gene.
3. Compare transcript-specific coordinates, length, composition, and charts.
4. Select the control again to collapse the group.

Isoform cards use transcript-level **Info** and **Charts** views. The canonical gene card retains the gene-wide information and literature actions.

4.10 Multi-gene alignment

Select **Aligned** when a loaded gene has at least two transcript tracks.

Controls include:

- **Gene to align** - chooses the multi-transcript gene;

- **Alignment mode** - Translated CDS, CDS nucleotide, or Full exon with UTRs;
- **Transcript order** - loaded order, reverse order, or transcript label;
- **Min. block identity** - hides ribbons below the selected threshold.

Translated CDS is the preferred starting point for protein-coding isoforms. CDS nucleotide keeps the coding comparison in nucleotide space. Full exon includes UTR extent and is useful for transcript-architecture differences.

4.11 Multi-gene LASTZ and MultiPIP

These modes compare transcripts from the selected gene using the canonical transcript as reference whenever it is available.

Configure:

- the gene to align;
- the alignment window: gene body, or the gene plus 5, 10, or 25 kb;
- minimum identity;
- minimum block or segment size;
- **Run local alignments**.

Use LASTZ Blocks to inspect discrete local matches. Use MultiPIP to inspect a reference-centered conservation profile. Export either view as SVG.

5. Cross-Species Gene Search

5.1 When to use it

Use Cross-Species Gene Search to compare a selected gene across organisms. Typical questions include:

- whether a gene is represented in several selected annotations;
- how representative transcript structures differ;
- which exon-level relationships are conserved;
- where local genomic sequence remains similar;
- how gene metrics vary across organism tracks.

5.2 Build the organism set

1. Select **Cross-Species Gene Search**.
2. Choose **Preloaded organisms**, **NCBI Search**, **Upload files**, or **Mixed sources**.
3. Add at least two organisms.
4. Confirm each assembly and source.
5. For uploads, verify annotation-genome pairing.

The organism selector allows multiple entries. In Desktop, only installed references appear as preloaded selections.

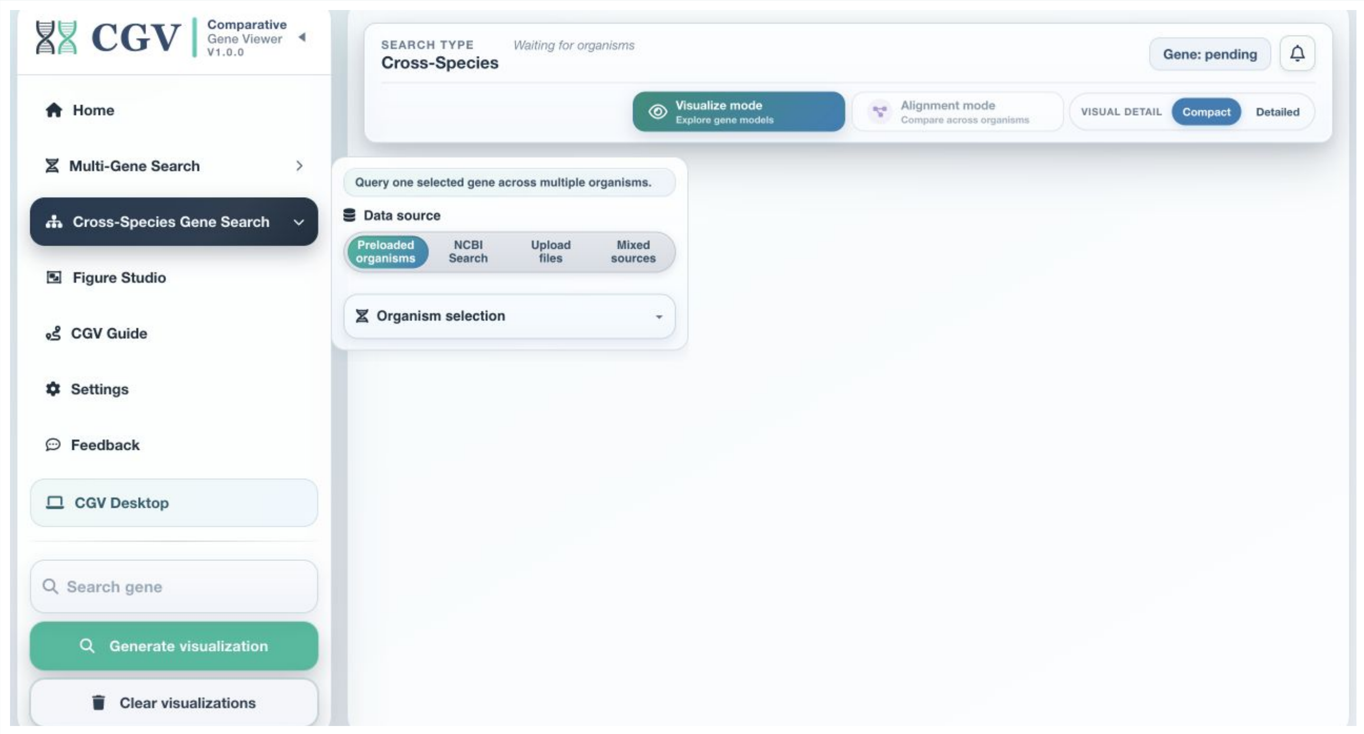


Figure 5.1. *Cross-Species Gene Search before execution. The source selector includes Preloaded organisms, NCBI Search, Upload files, and Mixed sources.*

5.3 Search one target gene

1. Enter one gene symbol, stable identifier, or known alias.
2. Select an autocomplete suggestion when available.
3. Select **Generate visualization**.

CGV checks each organism independently. A direct local match is used first. If no local match is found, enabled alias services can resolve alternate nomenclature in the background.

The context header marks organism states such as:

- local match;
- match recovered through external alias evidence;
- external search in progress;
- no match in the current annotation.

Select **About results** in the Cross-Species context header to review the comparison scope. CGV displays the selected gene when the same name or a resolved alias is available in at least two selected organisms. Additional family members found only within one organism are not added to this result; use Multi-Gene Search to explore those genes within that organism.

5.4 Interpret partial availability

Cross-species results may contain fewer cards than selected organisms. Common reasons include:

- the gene is absent from the annotation;
- the organism uses a different symbol or stable identifier;
- the assembly or annotation version differs from the expected record;
- no transcript structure is available;
- external alias evidence is insufficient or ambiguous.

Use the notification history and organism status pills to distinguish these cases. A missing card should be reported as "not resolved in the selected source" rather than as biological absence unless the source annotation has been independently verified.

5.5 Resolve suggestions

If no exact cross-species match is available, CGV can suggest similar genes shared by the selected annotations. Select one exact suggestion to rerun the comparison. Cross-species results require a usable match in at least two organisms.

If nomenclature differs strongly across species, select the external-alias option and review the resulting evidence badges.

5.6 Compact and Detailed views

Compact view is appropriate for a first pass across many organisms. Detailed view is appropriate for feature-level inspection.

Before alignment:

1. verify the representative gene and transcript label for each organism;
2. check strand, coordinates, and chromosome;
3. inspect expanded isoforms where they affect interpretation;
4. identify any source or alias badges;
5. confirm that genome sequence is available for alignment-oriented work.

5.7 Comparative Aligned Synteny

Aligned synteny compares transcript structures in a common left-to-right 5' to 3' display.

Controls include:

- **Alignment mode** - Translated CDS, CDS nucleotide, or Full exon with UTRs;
- alignment anchor/reference selection;
- **Track order** - biological similarity chain, load order, organism name, reverse order, or manual drag order;
- **Min. block identity**.

Ribbons indicate structural correspondence between adjacent displayed tracks. Hover a ribbon to review identity, support length, GC context, and mapping-event details.

5.8 LASTZ Block View

LASTZ Block View aligns every selected query locus against a reference locus.

Controls include:

- reference organism or track;
- **Alignment window** - gene body, or gene plus 5, 10, 25, or 50 kb;
- track order;
- minimum block length: 50, 100, 250, or 500 bp;
- minimum local identity;
- **Run local alignments**;
- **Suggest best reference**.

The full genomic window is used, including introns and selected flanking sequence. Conserved blocks outside the transcribed region can therefore represent non-coding regulatory context, but they require independent biological validation.

5.9 MultiPIP-style View

MultiPIP presents reference-centered conservation across the organism set. Configure:

- reference track;
- alignment window;
- track order;
- minimum segment length: 10, 25, 50, or 100 bp;
- minimum local identity;

- **Run local alignments**;
- **Suggest best reference**.

Use the view to identify conserved intervals that recur across several organisms. It summarizes local alignment support; it is not a multiple-sequence alignment for phylogenetic inference.

5.10 Reference selection

Choose a biologically meaningful, well-annotated reference with a complete genome sequence. **Suggest best reference** evaluates the loaded tracks and selects a strong reference candidate. The choice affects projection and visual order, so record it with exported results.

5.11 Sort cross-species cards

Cards can be sorted by:

- load order;
- total exon difference;
- transcript length;
- exon count;
- organism name;
- transcript name.

Use organism-name sorting for a predictable report layout. Use exon-difference or transcript-length sorting to foreground structural outliers.

6. Reading a result card

6.1 Card header

The header identifies the current organism, gene, chromosome, and transcript context. It can also show chromosome mapping or alias evidence.

Header controls include:

Control	Action
Function	Opens a gene-function summary from local annotation or NCBI Gene
Network	Opens the interactive STRING protein-interaction network
GO	Opens local GO annotations and supported online fallback
NCBI	Opens the exact GeneID or an organism-aware NCBI Gene search
Ensembl	Opens the relevant Ensembl portal search, including plant or fungal portals
UniProt	Opens a gene, protein-name, and organism-aware UniProt search
Alias	Displays the local or external alias evidence used to resolve the query
SVG	Exports the current structural plot as editable SVG
Download FASTA	Opens the Gene, Transcript, CDS, and Introns sequence menu
Close icon	Removes the current card after confirmation when enabled

External links open outside the CGV application. In Desktop they open in the default system browser.

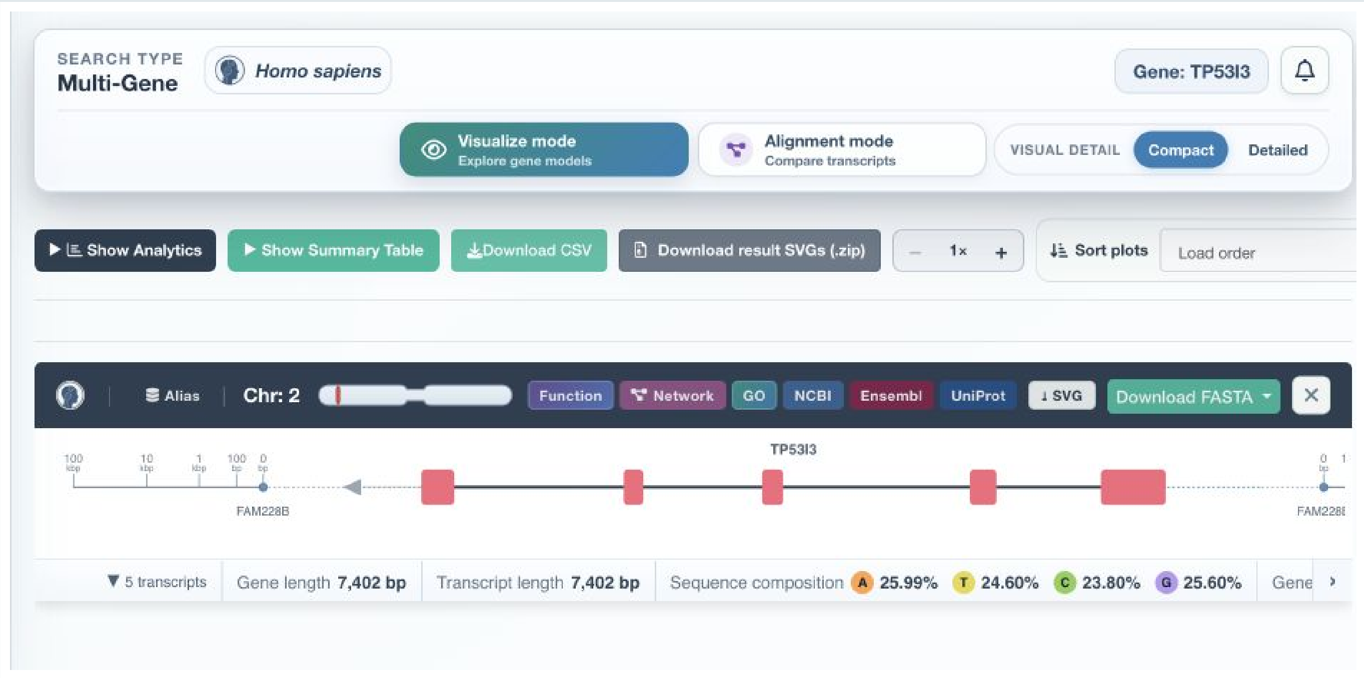


Figure 6.1. A real Compact result card for TP53I3 in Homo sapiens. The toolbar, database actions, structural model, transcript count, lengths, and sequence composition are visible.

6.2 Interactive gene model

The central model may contain:

- a gene or transcript backbone;
- exon blocks;
- CDS blocks;
- UTR features;
- inferred intron intervals;
- other annotated RNA or transcript features;
- strand-direction arrows;
- chromosome coordinates;
- neighboring genes and distances;
- promoter-side connector.

Hover over an interactive feature to inspect its type, coordinates, length, identifiers, attributes, and GC content where sequence is available.

Select a neighboring-gene marker or overlap band to open its context popup. The popup identifies the relation to the current gene, genomic coordinates, and strand. In Multi-Gene Search, **Visualize [gene] below** adds that neighbor as a new result card. In Cross-Species Search, neighbors remain genomic context only; adding one as a result is intentionally available only through Multi-Gene Search.

6.3 Compact and Detailed feature behavior

Compact view reduces the number of visible layers and uses combined structural regions. Detailed view displays more annotation layers and attribute-rich tooltips.

Use Detailed view before concluding that a region is coding or untranslated. A visually similar block can represent a different feature type in a source with incomplete or nonstandard annotation.

6.4 Strand and orientation

The central transcript is displayed in its biological orientation. Neighbor context and promoter direction follow the annotated strand. A negative-strand gene has its upstream promoter region on the higher-coordinate side.

Alignment modes normalize tracks into left-to-right transcript order. Their x-axis is comparative transcript space, not raw genomic coordinates.

6.5 Genomic context scale, neighbors, and overlaps

The solid center of the **Scale** is a linear chromosome or sequence axis for the displayed gene interval. Its labels use bp, kb, or Mb according to the span. Hover the line, ticks, or labels to inspect exact positions.

The dashed side sections place the nearest flanking genes on a compressed distance scale. Break marks separate these logarithmically compressed side sections from the linear gene interval. The side ticks show 100 bp, 1 kb, 10 kb, and at least 100 kb reference distances.

Overlapping genes are drawn as bands across the locus. Multiple overlaps can use separate lanes and a combined label. Use the marker popup or tooltip for the exact relation and overlap length; do not estimate a side distance from screen width because the neighboring-gene sections are compressed.

Use **Scale**, **Neighbors**, and **Overlaps** in the result context header to show or hide these layers independently. Hiding a layer does not change the annotation, coordinates, summary, or analysis.

6.6 Promoter region

The dashed promoter-side connector is interactive.

1. Select the promoter region.
2. Enter or slide to the required upstream length.
3. Choose a value from 100 to 5,000 bp in 10 bp increments. The default is 1,000 bp.
4. Select **Download promoter region**.

CGV extracts the strand-aware upstream interval and downloads it as FASTA. Coordinates are clipped to the sequence boundary when necessary.

NOTE	"Promoter region" in this tool means a configurable upstream sequence window. It does not assert experimentally validated promoter activity.

6.7 Card footer

The footer can be horizontally scrolled and provides:

- transcript count and isoform expansion;
- gene and transcript lengths;
- nucleotide composition;
- gene or transcript coordinates;
- **Info**;
- **Charts**;
- **Literature** on canonical gene cards.

Footer arrows appear when the available width cannot show all items.

6.8 Info popup

On a canonical gene card, **Info** opens gene statistics. On an isoform card it opens transcript information. Depending on the record, sections can include:

- gene summary;
- current transcript;

- architecture and feature counts;
- sequence metrics;
- neighboring-gene context;
- coordinate and strand information.

6.9 Charts popup

Canonical gene cards provide gene and isoform chart groups. Transcript cards provide transcript-oriented charts.

Transcript charts include:

- structure composition;
- intron map;
- nucleotide composition.

Gene and isoform charts include:

- transcript isoform length comparison;
- gene architecture by isoform;
- exonic and intronic base-pair comparison;
- sequence composition;
- gene-versus-transcript context;
- exon-length distribution;
- intron-length distribution.

Each chart has an information control and SVG export.

6.10 Removing results

Use the close icon on a card to remove only that result. Use **Clear visualizations** in the sidebar to remove all active cards. When **Confirm before deleting** is enabled, CGV asks before destructive workspace actions.

Removing a card changes the summary table, analytics set, and Figure Studio availability. Export or save the session first if the result may be needed again.

7. Alignment and conservation views

7.1 Select the view that answers the question

View	Best for	Coordinate basis	Primary evidence
Compact	Rapid structural overview	Genomic/transcript model	Annotation
Detailed	Feature-level inspection	Genomic/transcript model	Annotation
Aligned synteny	Structural correspondence	Normalized 5' to 3' transcript space	Annotation and aligned blocks
LASTZ Blocks	Discrete local sequence similarity	Reference genomic window	Local sequence alignment
MultiPIP	Reference-centered conservation pattern	Reference genomic window	Filtered local alignment segments

7.2 Aligned-synteny mapping events

CGV uses visual event categories:

- **1:1 direct** - one block corresponds preferentially to one block;
- **1:n or n:1 correspondence** - one block maps to several blocks, or several blocks map to one;
- **partial or ambiguous** - support is divided or does not form a clean direct mapping.

The ribbon tooltip is the authoritative source for the event label, identity, and support. Ribbon width and curvature are visual aids and should not be treated as independent quantitative measures.

7.3 Alignment modes

Translated CDS compares coding blocks in protein-oriented space when CDS features are available. It is robust to synonymous nucleotide differences and is the recommended first choice for protein-coding genes.

CDS nucleotide uses coding-sequence nucleotides. It is useful for closely related coding sequences or when nucleotide-level differences matter.

Full exon (+ UTRs) includes exon geometry and untranslated extent. Use it for transcript-architecture questions and UTR differences.

When a transcript lacks CDS, CGV can fall back to exon geometry so the track remains interpretable. Report such tracks separately from complete protein-coding comparisons.

7.4 Identity thresholds

Increasing the minimum identity removes weaker ribbons, blocks, or segments from view. It does not change the annotation.

- Use a low threshold for exploratory comparison across distant organisms.
- Use a moderate threshold to reduce visual noise.
- Use a high threshold when the question specifically concerns strong conservation.

Always report the threshold used. A visually empty track at a high threshold does not imply that no relationship exists.

7.5 Alignment windows

Gene body only restricts local alignment to the transcribed locus window.

Flanking-window options add sequence on both sides of the gene. They are useful for:

- promoter and proximal-regulatory comparison;
- identifying conserved non-coding intervals;
- observing boundary-spanning blocks.

Larger windows increase runtime and the number of candidate local matches.

7.6 Track order

Track order affects which adjacent tracks are connected in a ribbon view and how quickly a reader can recognize patterns.

- **Biological chain** places similar tracks near one another.
- **Loaded order** preserves the original selection order.
- **Organism A-Z** produces a stable reference layout.
- **Reverse current** reverses the current display.
- **Manual** allows drag-based arrangement.

Track order is a presentation choice unless the view explicitly recalculates adjacent correspondence. Record non-default ordering with the exported figure.

7.7 Local alignment execution

Select **Run local alignments** after changing reference, window, or other parameters. CGV reports the run state and raw hit count in the view footer.

Because LASTZ is computationally intensive, the notification popup opens automatically during LASTZ Blocks, MultiPIP, and reference-scoring runs. It shows progress and elapsed time. Keep CGV open and wait for the completion message before changing the source or closing Desktop.

If the local engine reports unavailable sequence:

- confirm that each organism has a genome file;
- confirm that annotation sequence identifiers match the genome;
- verify that the selected coordinates exist;

- reduce the organism set to isolate the problematic source.

If CGV reports a safety timeout or a locus window that is too large, select a smaller alignment window and retry. Completed results remain available even when another query track fails.

7.8 Sequence export from alignment views

Aligned-synteny, LASTZ, and MultiPIP cards provide **Download sequences**. The export contains the reference and displayed query sequences for the active comparison context. Use the exported headers to preserve organism, gene, sequence identifier, interval, and role.

7.9 Interpretation limits

Alignment views provide evidence of structural or local sequence similarity. They do not, by themselves, establish orthology, conserved regulation, functional equivalence, evolutionary direction, or statistical significance.

Combine CGV results with curated orthology, phylogenetic analysis, experimental evidence, and source-specific annotation review where those conclusions are required.

8. Summary tables and analytics

8.1 Summary table

Select **Show Summary Table** to inspect a searchable, sortable table behind the current cards. The table includes record labels and metrics such as:

- organism;
- gene and transcript identifiers;
- chromosome and strand;
- gene and transcript coordinates;
- gene and transcript length;
- exon and intron counts;
- exonic, intronic, CDS, and UTR measures;
- sequence composition and GC percentage;
- neighboring-gene distances where available.

Select **Download CSV** to preserve the current summary. The CSV is the preferred source for exact numeric reporting.

8.2 Opening analytics

Select **Show Analytics** after generating results. Analytics use the currently active card set. Removing cards or changing sources changes the comparison population.

Every analytics tab provides:

- an interactive chart;
- hover details;
- an information control explaining axes and interpretation;
- a chart-specific ordering selector;
- single-chart SVG export.

The **SVG ZIP** action exports the available analytics charts as a batch.

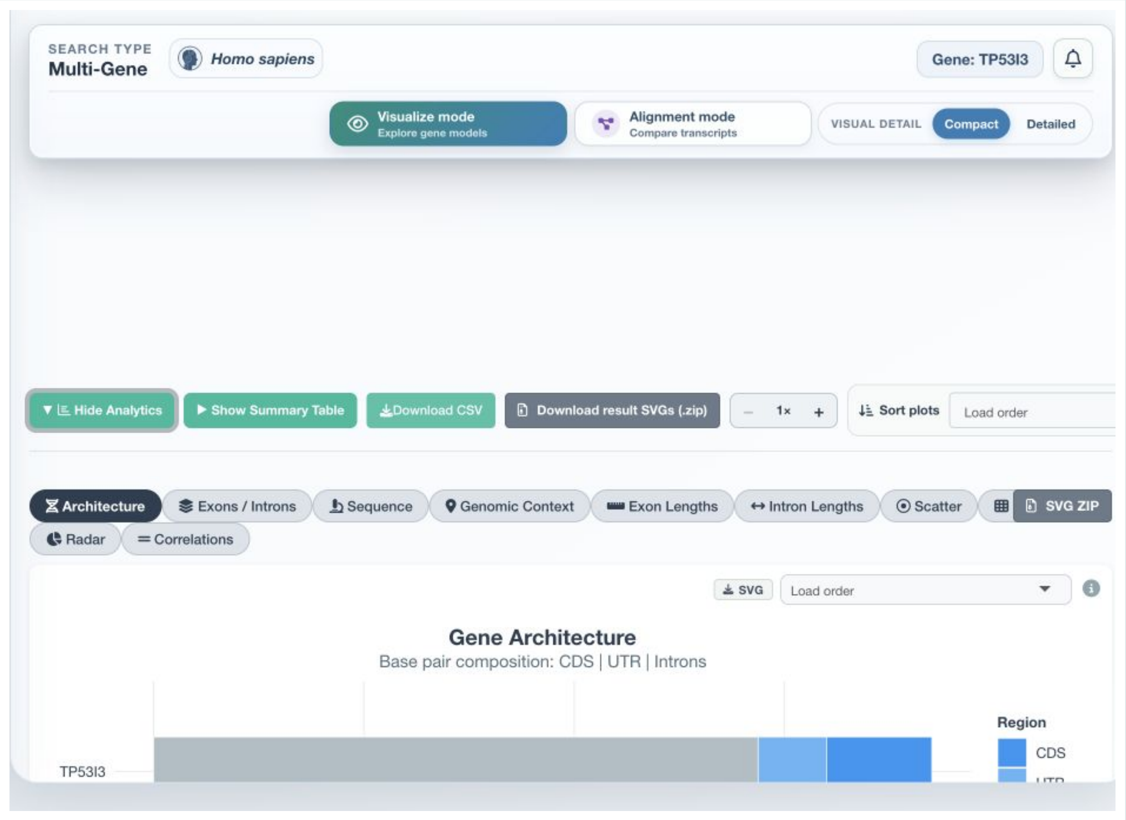


Figure 8.1. The analytics workspace with the ten chart families visible and Gene Architecture selected.

8.3 Gene Architecture

The stacked bar displays CDS, UTR, and intronic base-pair composition.

Use it to compare:

- total gene size;
- coding proportion;
- UTR contribution;
- intronic contribution.

Longer bars indicate larger genes. Similar total length can conceal different structural composition, so inspect the segments and tooltip values.

8.4 Exons / Introns

This view compares exon and intron counts together with exonic and intronic base pairs.

Use it to distinguish:

- many short features from a few long features;
- structurally compact genes from intron-rich genes;
- count differences from length differences.

Feature count is annotation-dependent and should not be interpreted as regulatory complexity without additional evidence.

8.5 Sequence

The sequence chart shows A, T, C, and G percentages and reports GC percentage.

Use it to compare broad nucleotide composition. N/A means the corresponding genome sequence could not be extracted; it does not represent zero content.

8.6 Genomic Context

The context chart shows edge-to-edge distances to the nearest upstream and downstream genes.

- Negative values indicate overlap.
- Zero or positive values indicate adjacency or a gap.
- Upstream and downstream panels define biological side; do not infer side from sign alone.

8.7 Exon Lengths

The exon distribution uses violin, box, and jitter layers on a log10 scale.

- the box represents the interquartile range;
- the line represents the median;
- dots represent individual exons;
- the violin represents the distribution shape.

8.8 Intron Lengths

The intron distribution shows inferred gaps between consecutive annotated exons. Single-exon transcripts are omitted. When no displayed transcript has introns, the chart provides an explicit no-intron message.

8.9 Scatter

The scatter plot compares GC percentage with gene length. Point encoding also supports interpretation of additional structural metrics such as exon count.

Use hover labels rather than estimating exact values from position. The chart is descriptive and does not imply causation.

8.10 Heatmap

The heatmap compares normalized gene metrics across the loaded set. It requires at least two usable records.

Values are scaled within the loaded comparison. A high color intensity means high relative to this set, not high relative to the organism or genome.

8.11 Radar

The radar chart compares six normalized metrics on a 0 to 1 scale. It is useful for visual profile comparison when at least two records have complete metrics.

Radar area should not be treated as a formal composite score. Compare individual axes and consult the summary table.

8.12 Correlations

The correlation matrix summarizes pairwise metric relationships. Base-pair and count metrics can be log10 transformed before correlation. Ordering options include default, alphabetical, mean absolute correlation, and clustered order.

Correlation requires enough complete records to be meaningful. Small gene sets can produce unstable or visually strong coefficients.

8.13 Analytics ordering

Ordering menus include load order, numeric high-to-low or low-to-high options, and label order. Ordering affects chart presentation only. For reproducible figures, keep the order visible in the exported caption or record it in the analysis notes.

8.14 Missing values

Analytics exclude or mark records that lack the required metric. Common causes are:

- unavailable genome sequence;
- incomplete transcript or exon annotation;
- no introns;
- no usable neighboring-gene record;

- insufficient records for a comparative statistic.

Do not replace N/A with zero unless the biological and computational meaning supports that decision.

9. Functional and contextual tools

9.1 Gene Function Summary

Select **Function** on a result card.

CGV first uses descriptions, notes, or products in the local annotation. When a local summary is unavailable and a resolvable NCBI Gene record exists, it retrieves and caches the NCBI Gene summary.

The dialog identifies:

- gene and transcript;
- chromosome;
- organism and TaxID;
- GeneID when available;
- annotation file;
- information source;
- search workflow.

Use the source label when citing or auditing the summary.

9.2 STRING Protein Network

Select **Network** to open an interactive protein-protein interaction network.

Visual roles are:

- red - target gene;
- orange - genes already displayed in CGV;
- gray - additional STRING interactors.

Edge thickness reflects STRING combined confidence. Hover nodes and edges for details. Drag nodes, pan, use wheel zoom, navigation controls, keyboard controls, or double-click to refit the network.

Select **SVG** to export the current network.

NOTE	STRING integrates several evidence channels, including experiments, curated databases, co-expression, text mining, genomic neighborhood, fusion, and co-occurrence. An edge is evidence of association, not necessarily direct physical binding.

9.3 Gene Ontology

Select **GO** to search GO annotations for the current gene and organism.

CGV groups results into:

- Biological Process;
- Molecular Function;
- Cellular Component.

Local GAF data are used when available. If no local entry is resolved, select the online lookup action to query the supported MyGene.info fallback. The popup reports the source and can reveal additional terms with **Show more**.

GO terms describe curated or inferred annotations from the source. Check evidence codes and source records when evidence strength matters.

9.4 Scientific Literature

Select **Literature** on a canonical gene card. CGV searches Europe PMC using the gene, known synonyms, organism, and organism aliases.

The literature window supports:

- sorting by most cited, newest, oldest, or relevance;
- client-side filtering by author, title, or journal;
- paged results;
- titles, authors, year, journal, abstract, citation count, and external record links.

Review the search context before treating the result list as exhaustive. Gene symbols that are common words or reused across taxa may need manual verification.

9.5 External database links

CGV builds organism-aware links:

- NCBI uses a GeneID when available, otherwise a symbol-and-organism search;
- Ensembl selects the main, plant, fungal, protist, metazoan, or bacterial portal from organism context;
- UniProt combines gene symbol, protein product, and organism context.

The external site is responsible for the record shown after the link opens. Confirm that its assembly, species, and identifier match the CGV card.

9.6 Assembly details

Select the information icon next to the preloaded organism selection or the organism pill in the result context.

Available details include:

- assembly name and description;
- organism and TaxID;
- BioSample and BioProject;
- submitter and release date;
- assembly type, level, and representation;
- RefSeq and GenBank accessions;
- assembly statistics;
- chromosome-name mapping;
- organism photographs and attribution when available.

Use assembly details to resolve chromosome aliases and to document the exact reference used.

9.7 Alias evidence

The **Alias** badge explains how the input query reached the plotted local record. It can show:

- input query;
- alias used;
- matched local symbol;
- local stable identifier;
- locally indexed aliases and their source database;
- alternate query variants reviewed.

This evidence is especially important when comparing species whose official nomenclature differs.

10. Figure Studio

10.1 Purpose

Figure Studio assembles independent CGV visualizations into a publication-ready multi-panel figure. It uses the SVG source of existing results and analytics, so the final composition remains vector-based when exported as SVG.

Open **Figure Studio** after generating the result charts that the figure requires.

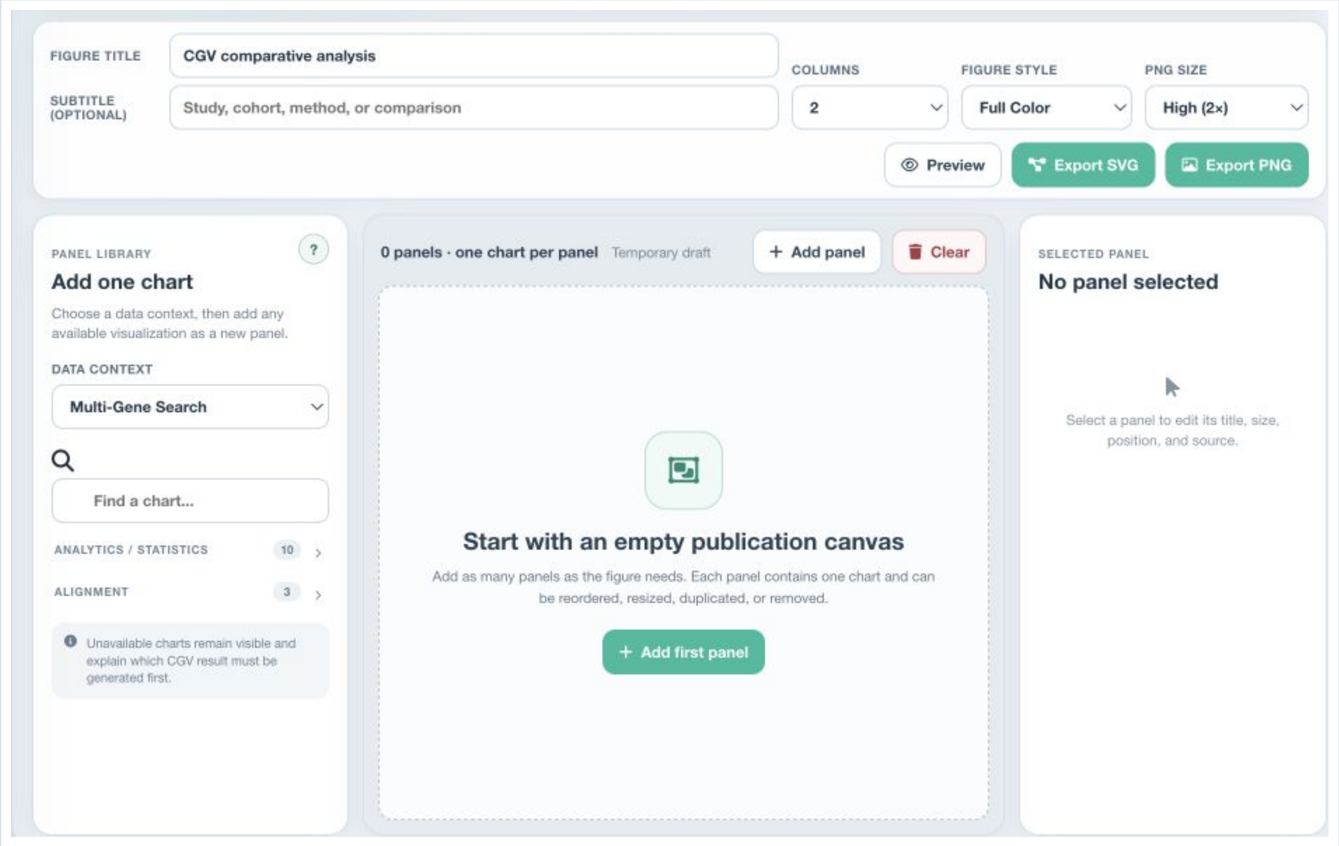


Figure 10.1. Figure Studio publication controls, panel library, empty canvas, and selected-panel inspector.

10.2 Publication toolbar

Configure:

- **Figure title**;
- optional subtitle;
- one, two, or three columns;
- **Figure style**;
- PNG size;
- preview and export.

Figure styles are:

- Full Color;
- Paper Color;
- Colorblind;
- Paper Gray;
- Paper Mono.

PNG size options are Screen, High, and Ultra. For editable publication work, SVG is the preferred master export.

10.3 Panel library

Choose a data context:

- Multi-Gene Search;
- Cross-Species Search.

Search the catalog by chart name. Available sources include:

- Gene architecture;
- Exons / introns;
- Sequence composition;
- Genomic context;
- Exon lengths;
- Intron lengths;
- GC vs gene length;
- Comparative heatmap;
- Comparative radar;
- Metric correlations;
- Aligned synteny;
- LASTZ blocks;
- MultiPIP;
- individual gene and transcript result plots.

Unavailable entries remain visible and explain which result or alignment must be generated first.

10.4 Add a panel

1. Select the required data context.
2. Locate a chart in the panel library.
3. Select **Add**.
4. Repeat for every panel.

NOTE	Each selection creates one independent panel. The same source may be added more than once when different placement or labeling is required.

10.5 Edit a panel

Select a panel on the canvas. The inspector provides:

- panel title;
- chart-source readout;
- width of one, two, or three columns;
- Auto, Compact, Standard, or Tall height;
- show or hide panel label and title;
- move Earlier or Later;
- duplicate;
- remove.

Auto height preserves the source chart's proportions and adapts to the number of genes or transcripts. Use manual height only when deliberate compression improves the composition without reducing legibility.

10.6 Reorder and resize

Drag a panel to reorder it, or use **Earlier** and **Later**. Panel width is constrained by the selected figure-column count. If the figure changes from three columns to two, spans are reduced to fit the new grid.

10.7 Undo, redo, new, and clear

- **Undo** reverses the most recent studio edit.
- **Redo** reapplies the reversed edit.
- **New figure** resets title, subtitle, layout, style, and panels after confirmation.
- **Clear** removes every panel but leaves the underlying CGV search results intact.
- **Back to results** returns to the scientific result page.

10.8 Preview

Select **Preview** to render the same composition used by export. Inspect:

- title and subtitle;
- panel labels;
- font and line legibility;
- panel order;
- empty or clipped areas;
- color profile;
- overall dimensions.

Export directly from the preview when the composition is correct.

10.9 Export

Export SVG creates the editable vector master with source metadata. **Export PNG** rasterizes the same composition at the selected scale. Very large figures are automatically constrained to a safe pixel count.

Keep the SVG even when a journal requests PNG or another raster format. It preserves editable text, paths, and panel structure.

10.10 Save the studio draft

The Figure Studio draft is temporary until the CGV work session is exported. Saving a session records:

- figure title and subtitle;
- column count;
- style and PNG scale;
- panel order, size, and labels;
- source references.

When the session is restored, CGV reconnects panels to their source results. If a source is missing, regenerate or restore that result and refresh the panel.

11. Exporting and preserving work

Need	Best export	Why
Edit a scientific figure	SVG	Vector structure, editable labels, scalable geometry
Share a fixed raster preview	PNG	Predictable appearance and convenient placement
Audit exact comparison values	CSV	Tabular data for statistics and record keeping
Continue sequence analysis	FASTA	Gene, transcript, CDS, intron, promoter, or alignment sequences
Resume the complete workspace	RDS	Sources, cards, views, settings, and Figure Studio state

11.1 Structural SVG

Use **SVG** in a result-card header to export one gene or transcript plot. SVG retains vector geometry and is appropriate for Inkscape, Illustrator, Affinity Designer, and compatible journal workflows.

11.2 Result SVG ZIP

Select **Download result SVGs (.zip)** to export all active structural plots in the current workflow. Expand required isoforms before export when they must be included.

11.3 Analytics SVG and SVG ZIP

Every analytics chart has an SVG action. Use **SVG ZIP** to export all available analytics charts for the active workflow.

11.4 Summary CSV

Select **Download CSV** above the summary table. Use CSV for exact numeric values, audit trails, supplemental tables, and downstream statistical work.

11.5 Sequence FASTA

The **Download FASTA** menu on a result card provides:

- **Gene** - strand-aware complete gene interval;
- **Transcript** - current transcript sequence;
- **CDS** - coding sequence assembled from CDS features;
- **Introns** - intronic intervals inferred from consecutive exons.

If the required genome or coordinates are unavailable, the export is disabled or contains only the resolvable context.

11.6 Promoter FASTA

Select the promoter-side region in the plot, choose 100 to 5,000 bp, and download the strand-aware upstream sequence.

11.7 Alignment sequences

Aligned-synteny, LASTZ, and MultiPIP cards export the sequences represented by the active reference and query configuration.

11.8 Figure Studio SVG and PNG

Use SVG as the editable master and PNG for presentation, review, or submission systems that require raster output.

11.9 Work-session RDS

Go to **Settings** and select **Export current session (.rds)**.

The session records:

- active navigation and preferred workflow;
- source modes and organism selections;
- Compact, Detailed, or alignment modes;

- sorting;
- enabled external alias services;
- summary visibility;
- search status and source context;
- active plots, plot data, metrics, and sequence blobs;
- query and batch state;
- Figure Studio draft.

To restore:

1. Open **Settings**.
2. Choose the saved `.rds` file.
3. Select **Restore session**.
4. Review the restored source and organism context.

Restoring replaces current visualizations.

> **TIP** Save a session before changing organism, clearing results, or beginning a substantially different analysis.

11.10 Reproducible export set

Select the unobtrusive **Share** action in the active result header after generating at least one result. CGV prepares a versioned reproducibility package. After the report is ready, select **Generate / download ZIP** to create the archive containing:

- `analysis.json`, with CGV version, workflows, queries, organisms, source

provenance (assembly/annotation accession, version, source, and SHA-256 checksum), alias decisions, parameters, unresolved organisms, and completed alignment metadata;

- a human-readable `README.md` and `CHECKSUMS.sha256`;
- a portable schema-v2 CGV session;
- available summary CSV, completed LASTZ/MultiPIP TSV, and captured SVG files;
- FASTA only when private sequence inclusion is explicitly enabled.

Complete reference genomes are not copied into the package. CGV records their assembly and annotation provenance instead. When private sequences are excluded, restored visualizations remain available but sequence-dependent downloads may be disabled.

The captured SVG set follows the selected report detail mode described below. In CGV Web, enabling reader downloads publishes the ZIP with the secret report; when that option is off, the author can still generate a private local copy from the Share dialog.

11.11 Read-only interactive reports

On CGV Web, the header **Share** action also creates an immutable secret URL. Before publishing, choose:

- **Complete** or **Fast** report detail;
- when both workflows contain results, whether to include **Multi-Gene**,

Cross-Species, or both;

- whether to include private or uploaded sequence content;
- whether readers may download the reproducibility ZIP;
- whether CGV should run the compatible current LASTZ and MultiPIP comparisons

before capturing both alignment views;

- an expiry of 7, 14, or 30 days. The default is 7 days.

When reader downloads are disabled, CSV, FASTA, and ZIP files are not published with the secret report. Tables and figures already included in the read-only HTML remain visible.

Complete is the recommended mode. It generates structural views, Analytics charts, and eligible aligned-synteny views that have not yet been opened. Optional LASTZ/MultiPIP execution is available only in this mode.

Fast captures only views that are already rendered and idle. It does not activate hidden Analytics, structures, alignments, or synteny. The report records those intentional omissions so readers can distinguish them from capture failures.

The report gives the gene structure full page width and presents chromosome position as a compact location aid. It contains aligned synteny when available, completed LASTZ/MultiPIP results, external results used in the analysis, and Figure Studio when available. In Complete mode, CGV derives the complete analytics chart set and summary tables during publication even when the author never opened those panels. Readers can use tooltips, zoom, filters, table sorting, and collapsible sections. The report cannot run new searches, modify the analysis, or contact external databases.

When one report contains both workflows, every location, structure, synteny, alignment, analytics, and table block is labelled and separated as **Multi-Gene** or **Cross-Species**. Detailed sections start collapsed to keep large analyses navigable. Gene-structure results are grouped by gene and organism; each group initially shows its primary transcript and provides a selector for one specific transcript or all captured isoforms.

For Multi-Gene results, CGV captures one aligned-synteny view for every loaded gene that has more than one transcript. Each view is labelled with that gene instead of with the complete search list. Report preparation temporarily renders uncached views behind the Share dialog. CGV freezes a visual copy of the current application beneath the modal during this process, so the visible workspace does not turn blank or expose intermediate mode changes.

Report generation is one of CGV's most intensive processes and can take several minutes, especially in Complete mode or when LASTZ/MultiPIP is requested. Keep CGV and the Share dialog open until the final link or local-file message appears. The dialog reports the current preparation stage.

If the browser cannot capture an expected element, CGV lists it before publication. The author must either cancel or explicitly continue with that element excluded; capture failures are never omitted silently.

Anyone with the URL can view and copy visible information. The link is not indexed, but it must still be treated as a secret. Reports created in the current browser appear under **Settings > Shared analysis reports**, where they can be copied or revoked.

CGV Desktop does not upload analyses. It exports the same interactive report as a self-contained HTML file plus the reproducibility ZIP. Select **Build files**, then save **Download HTML** and **Generate / download ZIP** separately through the operating-system save dialogs.

The optional pre-publication LASTZ/MultiPIP action is tied to the current report generation and only completed results are included. Persistent background queues for unfinished LASTZ/MultiPIP jobs are not part of this release; durable cancel/retry/resume jobs remain planned for a future phase.

12. Settings

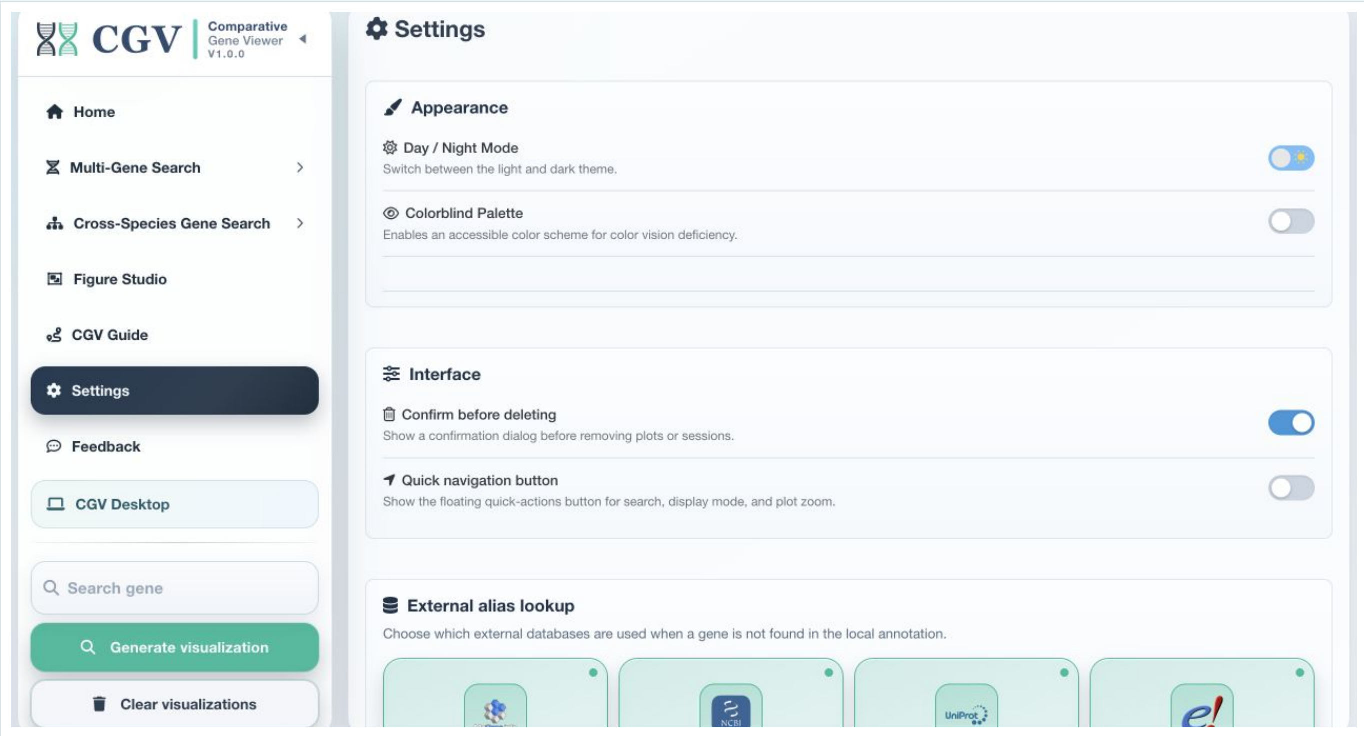


Figure 12.1. Settings provides appearance, interface, alias-source, organism, and work-session controls.

12.1 Day / Night Mode

The theme control switches between light and dark presentation. The selected theme is stored in the current application profile.

Theme changes affect on-screen plots and supported exported visualizations. Verify the final Figure Studio style independently of the interface theme.

12.2 Colorblind Palette

Enable **Colorblind Palette** to apply an accessible feature and chart color scheme. Use it for presentations, collaborative review, or publications where color-vision accessibility is required.

12.3 Confirm before deleting

When enabled, CGV asks before removing cards, clearing visualizations, or replacing destructive workspace state. Disable it only when rapid iterative clearing is more important than protection from accidental loss.

12.4 Quick navigation button

Enable this setting to display the floating search, display-mode, and zoom control described in Section 2.

12.5 External alias lookup

Select which services CGV may use when the local annotation does not contain the entered gene name:

- MyGene;
- NCBI Gene;
- UniProt;
- Ensembl.

Disabling a source narrows external resolution and can reduce background network activity. It can also reduce the number of recoverable aliases. Local exact matches and local alias-index matches remain preferred.

12.6 Organisms - Desktop only

The Organisms section manages locally installed references.

Controls include:

- installed, available, and pending counts;
- **Open catalog**;
- **Remove organisms**;
- **Refresh**;
- data and cache paths;
- catalog search;
- status filters: All, Available, Not installed, Installed, Updates.

Installed references appear in the preloaded selectors.

12.7 Work sessions

Use **Export current session (.rds)** and **Restore session** as described in Section 11. Session files are CGV workspace snapshots, not interchangeable biological data formats.

12.8 Shared analysis reports - Web

This section lists secret reports created in the current browser. Use it to open or copy a report URL, or to revoke the report before its scheduled expiry.

The list itself is stored only in the browser and does not require a CGV account. Clearing browser storage can remove the local receipt without revoking the published report; the secret link still expires automatically.

13. CGV Desktop

13.1 What Desktop adds

CGV Desktop packages the complete scientific interface with a local application runtime. It provides:

- a private localhost session inside the desktop window;
- bundled R and analysis tools;
- local LASTZ, sequence, and indexing utilities;
- persistent organism packages and caches;
- user-selected storage;
- local diagnostics;
- desktop update handling;
- the same search, analytics, Figure Studio, and export features as CGV Web;
- self-contained interactive HTML reports and reproducibility ZIPs saved only

to a location chosen by the user.

CGV Desktop does not require a separate R, Docker, WSL, or command-line installation for normal use.

13.2 Supported installers

Platform	Installer	Architecture
macOS	DMG	Apple Silicon arm64 and Intel x64
Linux	AppImage or DEB	x86_64
Windows	Assisted offline per-user NSIS setup	Windows 10 or 11 x64

Allow at least 2 GB for the application, plus storage for every organism installed from the catalog.

13.3 First launch

1. Open CGV Desktop.
2. On the first packaged Windows launch, choose a writable folder for genomes and caches when prompted. On macOS and Linux, CGV initializes its application-managed data location.
3. Allow the local runtime to initialize.
4. Wait for the main CGV interface.
5. Open **Settings** and install at least one organism.

The selected storage folder contains:

- `data` - genomes, annotations, manifests, and downloaded packages;
- `cache` - reusable indexes and derived cache data.

13.4 Organism catalog

1. Open **Settings**.
2. In **Organisms**, select **Open catalog**.
3. Search by organism name or filter by status.
4. Select **Download**.
5. Follow download, verification, extraction, and cache-preparation progress.
6. Return to the search workflow and select the installed organism.

Dataset states are:

- bundled;
- not installed;
- partial;
- installed;
- update available.

Downloads use temporary partial files, SHA-256 verification, safe ZIP extraction, and a local installation record. Existing verified files are reused. Active downloads can be canceled, and partial files are removed.

13.5 Verify and update an organism

Use **Verify** for an installed catalog entry. If a newer package is available, the catalog marks it as an update. Install the update from the same entry, then refresh the source selector.

13.6 Remove local organisms

Select **Remove organisms** to open the installed-organism list. Choose one, several, or **Select all installed organisms**, then confirm **Remove selected**. CGV removes only the selected catalog datasets and their associated local files and caches from the active desktop profile. This is a destructive data-management action. Export sessions and verify the storage path before confirming.

13.7 Change the data folder

On Windows, use **File > Change data folder...**.

CGV restarts and uses the newly selected folder. Existing genomes and caches are not moved or deleted. Select the previous folder again to reuse its content, or move the data manually while CGV is closed.

The Windows **File** menu also provides:

- **Open data folder**;
- **Show diagnostics log**;
- **Privacy policy**.

13.8 Automatic updates

Direct Windows releases check for updates, download them in the background, and install them when CGV restarts or quits as indicated by the update notification. Dataset updates remain separate from application updates and are managed in the organism catalog.

13.9 Diagnostics log

Startup and runtime details are kept out of the normal interface and written to:

Platform	Default log
macOS	`~/Library/Application Support/CGV Desktop/logs/startup.log`
Linux	`~/.config/CGV Desktop/logs/startup.log`
Windows	`%LOCALAPPDATA%\CGV Desktop\logs\startup.log`

Share the relevant final lines with a bug report. Review the file first if local paths are sensitive.

13.10 Desktop privacy model

The Shiny service listens on a private local port and is displayed inside Electron. Installed reference data and caches remain in the selected local storage folder.

External features still require network access, including:

- NCBI assembly search and download;
- external alias lookup;
- GO online fallback;
- Europe PMC literature;
- STRING networks;
- organism imagery;
- external database links;
- Feedback delivery and its optional confirmation email;
- application and dataset updates.

13.11 Uninstallation

Removing the desktop application preserves the user-selected storage folder and installed datasets. Delete that folder separately only when its genomes, annotations, caches, packages, and session-related data are no longer required.

14. CGV Guide

CGV Guide complements this manual with short visual demonstrations.

14.1 Guide routes

Desktop Downloads covers opening Settings, browsing the catalog, filtering, downloading, verifying availability, and selectively removing one, several, or all downloaded organisms.

Multi-Gene Search covers:

- preloaded, NCBI, and uploaded sources;
- single and batch gene entry;
- visualization generation;
- Compact and Detailed inspection;
- transcript alignment;
- figure and table export.

Cross-Species Search covers:

- preloaded, NCBI, uploaded, and mixed sources;
- one-gene search;
- visualization generation;
- Compact and Detailed inspection;
- aligned synteny, LASTZ, and MultiPIP;
- alignment export.

Common Analysis covers:

- analytics;
- summary tables;
- transcript variants;
- gene information;
- promoter sequences;
- literature;
- organism and assembly information;
- external alias configuration;
- save and load session;
- sharing an expiring read-only report URL in Web or exporting the interactive

HTML and ZIP locally in Desktop;

- clear visualizations.

Figure Studio covers:

- opening the workspace from the current analysis;
- adding structural, alignment, and analytics panels;
- arranging, resizing, labelling, and styling the composition;
- previewing and exporting SVG or PNG;
- including a non-empty Figure Studio composition in an interactive report.

14.2 Use the Guide effectively

1. Choose a route.
2. Select a numbered step.
3. Open a substep when available.
4. Watch the short demonstration.
5. Use the route action to open the corresponding CGV workflow.
6. Return to the Guide whenever a control sequence is unfamiliar.

The Guide is designed for procedural learning. Use this manual for complete option reference, interpretation guidance, and troubleshooting.

15. Troubleshooting

15.1 A gene is not found

Check, in order:

1. the correct workflow is active;
2. the intended organism or organisms are selected;
3. the query spelling and capitalization;
4. autocomplete suggestions;
5. stable IDs and known synonyms;
6. partial-name suggestions;
7. enabled external alias sources;
8. alias evidence from another successful record;
9. the source annotation version.

For Cross-Species Search, remember that at least two organism matches are required for a comparative result.

15.2 The wrong gene was resolved

- Review the **Alias** badge.
- Compare the official symbol, local ID, chromosome, and description.
- Rerun using the exact stable identifier.
- Disable unsuitable external alias sources.
- Check whether the symbol is reused by several gene families or taxa.

15.3 An upload pair is rejected

- Confirm one genome per annotation.
- Confirm both describe the same assembly.
- Compare chromosome or contig identifiers.
- Use clear matching file names.
- Decompress or convert unusual archives to an accepted format.
- Remove unrelated files from a multi-file selection.

15.4 A plot has N/A sequence values

- Verify that a genome file is available.
- Confirm the chromosome exists in the genome.
- Check annotation-genome build agreement.
- Check that coordinates fall inside the sequence.
- For an uploaded FASTA, verify headers match annotation sequence IDs.

15.5 Promoter download fails

- Confirm the plot has a genome sequence.
- Confirm transcript coordinates and strand.
- Reduce the upstream length near a chromosome boundary.
- Check sequence-identifier mapping in assembly details.

15.6 GO is empty

- Confirm the selected organism has a local GO file.
- Check that the GAF identifiers match the plotted record.
- Select the online lookup action.
- Try the official stable gene ID.
- Verify the organism TaxID.

15.7 STRING has no network

- Confirm the gene resolves to a protein-coding record.
- Verify the organism TaxID.
- Try an official symbol or stable ID.
- Check network access.
- Recognize that some organisms or proteins have limited STRING coverage.

15.8 Literature results are noisy

- Verify the organism shown in the dialog.
- Filter by author, title, or journal.
- Sort by relevance.
- Follow the external record and confirm the gene context.
- Search the official stable identifier separately for ambiguous symbols.

15.9 Alignment is empty

- Confirm at least two usable tracks.
- Confirm genome sequence for every track.
- Lower the identity threshold.
- Lower the minimum block or segment length.
- use a smaller, well-annotated reference set;
- select another reference;
- verify the alignment window and coordinates.

15.10 Alignment is slow

- Begin with gene body only.
- Reduce the organism or transcript set.
- Increase the minimum segment or block length.
- Avoid running multiple large NCBI downloads simultaneously.
- In Desktop, allow the first run to build reusable caches.
- Keep CGV open while the automatic status popup shows LASTZ or MultiPIP as active.
- If a safety timeout or oversized-window message appears, reduce the alignment window before retrying.

15.11 Analytics are missing

- Confirm at least one result card is active.

- Open the analytics section after plot rendering completes.
- For Heatmap, Radar, and Correlations, load at least two records with usable metrics.
- Confirm sequence-dependent metrics have genomes.
- Review the explicit no-data message in the chart.

15.12 SVG export is incomplete

- Wait for all cards or charts to finish rendering.
- Expand isoforms that must be exported.
- Use the specific card or analytics export control.
- Retry after switching back to the relevant result tab.
- Use Figure Studio preview to identify an unavailable source.

15.13 A session does not restore as expected

- Confirm the file is a CGV `.rds` work session.
- Confirm the file is accessible and not truncated.
- Restore into the same or a compatible CGV version.
- Keep original upload data available for operations that need fresh source access.
- Reinstall missing Desktop organisms.
- Regenerate any Figure Studio source marked unavailable.

15.14 Desktop does not start

1. Wait for runtime preparation to complete on first launch.
2. Confirm the storage folder is writable.
3. Open the diagnostics log.
4. Confirm free disk space.
5. Restart CGV Desktop.
6. On macOS, approve the application in **System Settings > Privacy & Security** when required.
7. On Linux, confirm AppImage execution permission or use the DEB package.
8. On Windows, use the signed per-user installer and review security prompts.

15.15 A Desktop organism does not appear

- Open **Settings > Organisms**.
- Refresh the catalog.
- Confirm the state is installed.
- Select **Verify**.
- Restart CGV Desktop if the selector was already open during installation.
- Confirm the data path belongs to the active profile.

15.16 Feedback and support

Open **Feedback** and choose **Report a Bug** or **Suggest a Feature**.

A useful bug report includes:

- short title;
- exact steps to reproduce;
- expected behavior;

- active workflow;
- organism and assembly;
- gene query;
- view and alignment settings;
- browser or operating system;
- CGV Desktop log excerpt when applicable.

Full name, a valid reply email, a short title, and a detailed description are required. CGV also records the active application section, submission time, and page context. The email address is used only for this submission and follow-up.

After successful delivery, CGV clears the submitted detail fields and displays a confirmation notice. When confirmation email is enabled, a copy with the submission reference is sent to the reporter; failure of that copy does not discard feedback already delivered to the CGV inbox.

If Desktop reports that the message was saved locally but cannot be sent, use the Feedback page at `cgv.mobilomics.org`, use `cgvapp.com` while the official server is unavailable, or email `cgvviewer@gmail.com`. Repeated or duplicate submissions can be temporarily rate-limited; follow the notification message before retrying.

Appendix A. Supported reference organisms

The curated registry contains the following 25 organism references. Availability in Web depends on the server deployment. In Desktop, installed catalog entries appear in the preloaded selectors.

Organism	TaxID	Kingdom	Reference assembly
Arabidopsis thaliana	3702	Plantae	GCF_000001735.4
Botrytis cinerea B05.10	332648	Fungi	GCF_000143535.2
Caenorhabditis elegans	6239	Animalia	GCF_000002985.6
Candida albicans	5476	Fungi	GCF_000182965.3
Canis lupus familiaris	9615	Animalia	GCF_011100685.1
Danio rerio	7955	Animalia	GCF_049306965.1
Desmodus rotundus	9430	Animalia	GCF_022682495.2
Drosophila melanogaster	7227	Animalia	GCF_000001215.4
Equus caballus	9796	Animalia	GCF_041296265.1
Fragaria vesca subsp. vesca	101020	Plantae	GCF_000184155.1
Homo sapiens	9606	Animalia	GCF_000001405.40
Malus domestica	3750	Plantae	GCF_042453785.1
Mus musculus	10090	Animalia	GCF_000001635.27
Neurospora crassa	367110	Fungi	GCF_000182925.2
Oryza sativa Indica Group	39946	Plantae	ASM465v1
Oryza sativa ssp. japonica	39947	Plantae	GCF_034140825.1
Pan troglodytes	9598	Animalia	GCF_028858775.2
Phaseolus vulgaris	3885	Plantae	GCF_000499845.2
Saccharomyces cerevisiae	4932	Fungi	GCF_000146045.2
Solanum lycopersicum	4081	Plantae	GCF_036512215.1
Sus scrofa	9823	Animalia	GCF_000003025.6
Brachypodium distachyon	15368	Plantae	GCF_000005505.3
Vitis vinifera	29760	Plantae	GCF_030704535.1
Xenopus laevis	8355	Animalia	GCF_017654675.1
Zea mays	4577	Plantae	GCF_902167145.1

Appendix B. Input and output reference

B.1 Input formats

Purpose	Formats
Annotation	GFF, GFF3, GTF, text
Genome	FASTA, FNA, gzip-compressed FASTA/FNA, 2bit
Assembly report	NCBI-style text
Assembly statistics	NCBI-style text
GO annotations	GAF, gzip-compressed GAF
Work session	CGV RDS

B.2 Output formats

Output	Format	Scope
Structural figure	SVG	One card
Structural figure bundle	ZIP of SVG	Active result set
Analytics chart	SVG	One analytics tab
Analytics bundle	ZIP of SVG	Active analytics set
Publication figure	SVG or PNG	Figure Studio composition
Summary table	CSV	Active result set
Gene, transcript, CDS, introns	FASTA	One card
Promoter window	FASTA	One selected promoter interval
Alignment sequence set	FASTA	Active alignment view
Work session	RDS	Complete CGV workspace

B.3 Annotation expectations

CGV works best when the annotation contains:

- gene features;
- transcript or mRNA features;
- exon features;
- CDS and UTR features where applicable;
- stable parent-child identifiers;
- valid sequence identifiers;
- numeric start and end coordinates;
- strand.

Fallback logic supports several transcript and non-coding RNA feature names, but nonstandard annotations may require validation before interpretation.

Appendix C. Keyboard and interaction reference

Interaction	Result
Enter in an active gene field	Runs the current single-gene search
Space in Multi-Gene batch input	Adds the current gene as a batch chip
Ctrl+Enter in Multi-Gene input	Adds the current gene as a batch chip
Escape	Closes the active supported popup or modal
Hover a feature	Shows annotation or metric details
Select a neighboring or overlapping gene	Opens relation, coordinate, and strand details
Select **Visualize [gene] below**	Adds that neighbor in Multi-Gene Search only
Select **Scale**, **Neighbors**, or **Overlaps**	Shows or hides that genomic-context layer
Select promoter connector	Opens promoter-length and FASTA controls
Drag Figure Studio panel	Reorders panels
Drag STRING node	Rearranges the network
Mouse wheel in STRING	Zooms the network
Double-click STRING canvas	Fits the network
Floating plus button	Opens quick search, display, and zoom actions

Appendix D. Glossary

Alias - An alternative gene symbol, stable identifier, synonym, or database-specific name used to resolve a local annotation record.

Annotation - A structured description of genomic features and their coordinates, typically stored as GFF3 or GTF.

Assembly - A specific version of an organism's reference genome sequence.

Canonical transcript - The primary transcript representative selected for a gene in the current annotation and CGV context.

CDS - Coding sequence translated into protein.

Exon - A transcribed feature retained in a mature transcript; coding and untranslated portions can both occur within exons.

FASTA - A text format for nucleotide or protein sequences.

GAF - Gene Association File used for Gene Ontology annotations.

Gene body - The interval covered by the annotated gene or selected transcript locus, excluding optional flanks.

GC percentage - The fraction of sequence bases that are G or C.

Intron - The interval between consecutive annotated exons in a transcript model.

LASTZ - A local sequence-alignment program used by CGV for block and conservation views.

MultiPIP - A reference-centered display of local conservation segments across several query tracks.

Orthology - An evolutionary relationship between genes separated by speciation. CGV provides comparative evidence but does not by itself establish orthology.

Preloaded organism - A reference whose annotation, genome, and registry metadata are ready for selection in the current CGV environment.

Promoter window - In CGV, a user-defined upstream sequence interval from 100 to 5,000 bp.

Synteny - Conserved genomic or feature order. CGV's aligned-synteny view focuses on gene and transcript structural correspondence.

Transcript isoform - One annotated transcript model among several associated with the same gene.

UTR - Untranslated region of a transcript.

Work session - A CGV RDS snapshot containing plots, settings, source context, and Figure Studio state.

Appendix E. Quick-reference checklists

E.1 Before running a search

- Correct workflow selected.
- Correct organism or organism set.
- Assembly version confirmed.
- Annotation and genome paired.
- Optional GO and assembly files added.
- Query spelling or stable ID checked.

E.2 Before interpreting a result

- Gene and transcript labels verified.
- Organism and chromosome verified.
- Alias evidence reviewed.
- Strand and coordinates checked.
- Missing sequence values identified.
- Isoforms expanded where relevant.
- Detailed view reviewed for feature-level claims.

E.3 Before interpreting an alignment

- At least two complete tracks.
- Reference recorded.
- Alignment mode recorded.
- Window recorded.
- Identity and length thresholds recorded.
- High-threshold empty results checked at a lower threshold.
- Conclusions separated from formal orthology or regulatory claims.

E.4 Before export

- Rendering complete.
- Desired cards and isoforms active.
- Sort order correct.
- Figure Studio preview checked.
- SVG master saved.
- CSV saved.
- Relevant FASTA saved.
- Session RDS saved.
- Source assembly and settings documented.

E.5 Before closing Desktop

- Session exported.
- Long downloads complete or intentionally canceled.

- Dataset state verified.
- Update status reviewed.
- Diagnostics captured if an error occurred.

Document control

Title: CGV User Manual - Comparative Gene Viewer

Coverage: CGV Web and CGV Desktop

Product version: 1.1.0

Manual revision: 30 July 2026

Primary application: https://cgv.mobilomics.org

Source repository: https://github.com/raulrojas22/CGV

This manual describes the complete user-facing behavior represented by CGV version 1.1.0, including the desktop data catalog, aligned-synteny tools, LASTZ and MultiPIP views, Figure Studio, interactive read-only reports, reproducibility exports, and session management.